

Deploy AI-powered visual inspection for aerospace component quality assurance, leveraging the Boeing-Odysight.ai partnership to validate defect detection in manufacturing.

Aerospace and defense industry × Automated quality inspection × Computer vision · OS123 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
44 is the world moving	25 can we play, can we win	MEDIUM 0 named in the evidence	€16.0m modelled evidence · per year	NEXT pilots published no volume pr...	L2 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Aerospace and defense industry, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity deploys AI-powered visual inspection for aerospace component quality assurance, leveraging the Boeing-Odysight.ai partnership to validate defect detection in manufacturing. It targets aerospace manufacturers who need to move from manual inspection to automated, real-time defect detection. The time is right because recent advances in computer vision and the Boeing-Odysight.ai agreement signal market readiness.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€19.4m €5.6m – €68.9m	€16.0m €4.6m – €56.8m	€40k €11k – €142k	modelled
Observed: tendered contract value, annualised	€45.7m	€45.7m	€114k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Aerospace and defense industry (EU27_2020)	3,354 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Computer vision	7.8 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TIR	2025
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	780 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

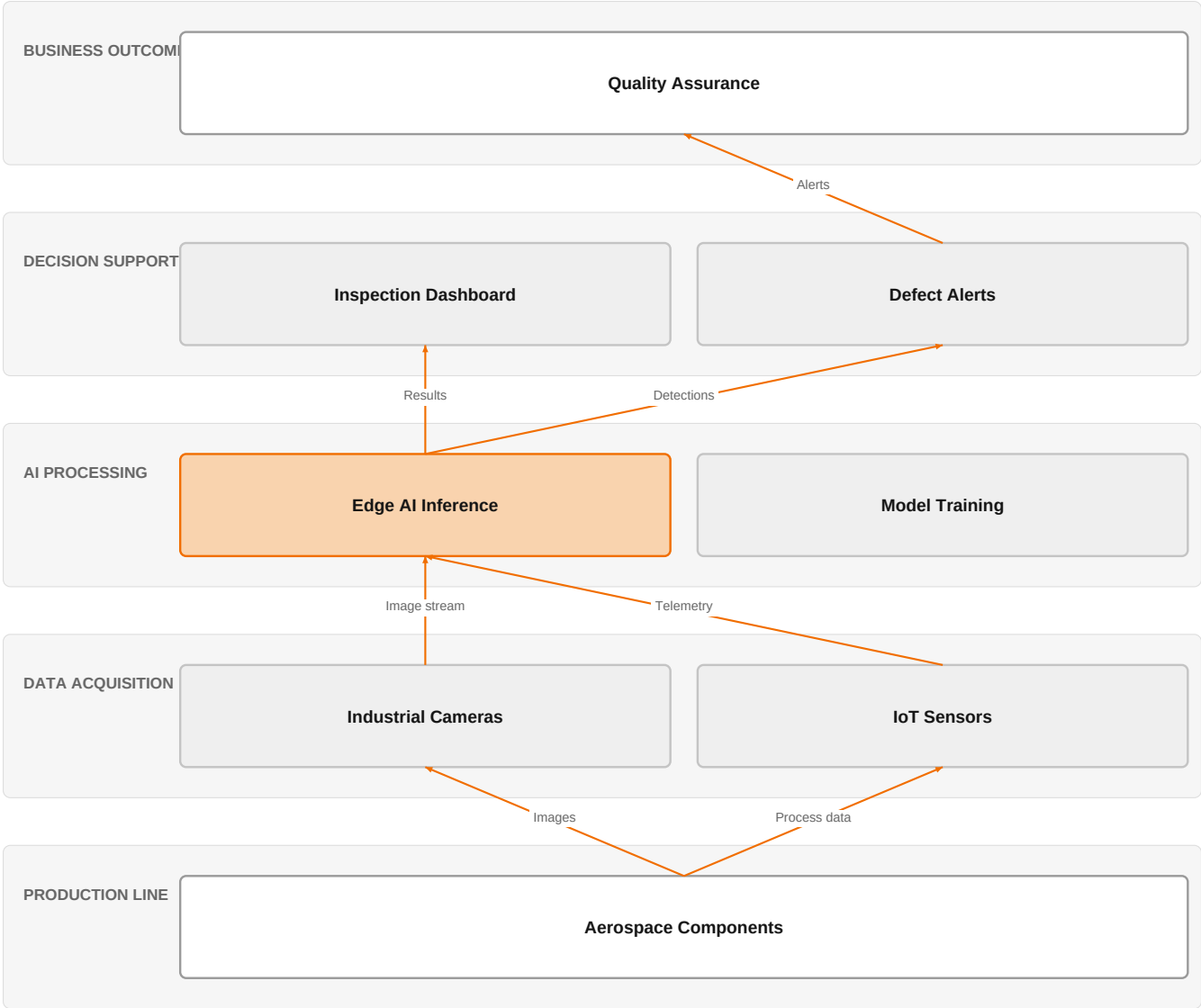
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€45.7m per year	observed	TED (Tenders Electronic Daily) · ted	2026-07-07..2026-07-08

Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a
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Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 3 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

AI Visual Inspection for Aerospace QA



Orange asset Partner Customer-owned Third party / to be sourced

This diagram shows how Orange integrates IoT and edge AI to deliver real-time defect detection for aerospace quality assurance.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The buyer is typically the head of manufacturing engineering or the quality assurance director at an aerospace component manufacturer. The pain is the high cost and unreliability of manual inspection, which leads to missed defects and rework. The trigger is often a quality audit finding or a customer complaint that forces a search for automated solutions. They need to demonstrate to regulators and customers that their quality assurance is robust and traceable.

Why it is hot now — every claim bound to a dated source

- Odysight.ai has signed an agreement with Boeing to demonstrate AI-powered capabilities, likely for visual inspection in aerospace. [SIG-D44EEFF68BEA]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts to deploy edge-based computer vision systems, leveraging NVIDIA's edge computing platforms for real-time inference. Orange's Defense and security experts would ensure the solution meets aerospace security and sovereignty requirements. Orange Group researchers would adapt state-of-the-art few-shot learning models to the customer's specific defect types. The engagement would start with a pilot on a single production line, using the Boeing-Odysight.ai partnership as a reference for validation, then scale across the plant.

Why Orange can win

Orange can win because it combines industrial IoT expertise with defense-grade security and sovereignty, which is critical for aerospace manufacturers. The partnership with NVIDIA provides access to proven edge AI hardware. However, the assets are thin: Orange does not have a proprietary computer vision model or a deep aerospace manufacturing track record, so the win will depend on the strength of the partner ecosystem and the credibility of the Boeing-Odysight.ai reference.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Computer vision	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 5.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Bosch and Honeywell, who have deep industrial OT expertise and sell computer vision solutions. HENSOLDT and DEDRON by Axon are security specialists with aerospace and defense presence. Atos/Eviden and Sopra Steria are global systems integrators active in aerospace and defense. Airbus Protect and Dassault Systemes/3DS Outscale are also active, with the latter offering sovereign cloud. The customer will compare Orange against these players on domain expertise, security, and integration capability.

Sources: SIG-D44EEFF68BEA manilatimes.net 2026-08-13

Competitor	Category	Position here	Basis	Evidence
Bosch	Industrial / OT specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
DEDRON by Axon	Security specialist	sells Computer vision; active in Aerospace and defense industry; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Computer vision; active in Aerospace and defense industry	structural	—
Honeywell	Industrial / OT specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	active in Aerospace and defense industry	structural	—
Sopra Steria	Global systems integrator	active in Aerospace and defense industry	structural	—
Airbus Protect	Security specialist	active in Aerospace and defense industry	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Aerospace and defense industry; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How many defects are currently missed by manual inspection, and what is the cost of rework or recalls?
- 2 Do you have a centralized data lake of inspection images, or are they siloed across plants?
- 3 What is your current process for validating new inspection technologies before deployment?
- 4 Are you subject to any specific regulatory requirements for traceability of quality assurance decisions?
- 5 Have you already evaluated any AI-based inspection solutions, and what were the outcomes?
- 6 What is your timeline for implementing automated inspection, and is there a specific production line you would pilot on?

Objections you will meet

They will say	You can answer
We already have a manual inspection process that works; why change?	Manual inspection is error-prone and costly. Recent studies show that computer vision can detect defects in real time, reducing missed defects and improving efficiency. The Boeing-Odysight.ai partnership demonstrates that major aerospace players are moving in this direction.
AI inspection is not proven for aerospace-grade components.	It's true that aerospace has stringent requirements. However, recent research on vision transformers and few-shot learning shows that AI can be adapted to new defect types with limited data. We would start with a pilot to validate on your specific components.
We are concerned about data security and sovereignty.	That's a valid concern. Orange brings defense and security expertise, and we can deploy on sovereign infrastructure if needed. We can also use federated learning to keep data on-premises.

Risks and unknowns

The main risk is that the Boeing-Odysight.ai partnership is only a demonstration, not a proven production system. The technology is still evolving, and few-shot learning may not generalize to all defect types. There is also uncertainty about the customer's willingness to share production data for model training, and about regulatory acceptance of AI-based inspection. Orange's lack of a proprietary computer vision model could be a disadvantage.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for an aerospace quality-inspection opportunity, outlining how Orange's IoT experts, defense/security expertise, and NVIDIA L2 partnership could deliver a computer vision solution, referencing the Boeing-Odysight.ai pilot as a validation point.
Sales	In a meeting with an aerospace manufacturer's head of quality, say: 'Given the recent Boeing-Odysight.ai pilot, we see strong potential for AI-driven visual inspection to reduce defects. Could we explore how our IoT and computer vision capabilities, with NVIDIA as a partner, might support your quality assurance?'
Strategist / Innovator	Commission a deep-dive brief on the Boeing-Odysight.ai visual inspection pilot, mapping how Orange's IoT and defense/security experts plus NVIDIA L2 partnership can support a comparable aerospace quality-inspection offer.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-8CA4FFD436BC	2026-03-27	openalex.org	1	Adapting Vision–Language Models for Few-Shot Industrial Defect Detection
SIG-0964410DC575	2025-12-17	openalex.org	1	Generative and Predictive AI for digital twin systems in manufacturing
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-3E11F4C71251	2025-11-13	openalex.org	1	Artificial Intelligence (AI) in Construction Safety: A Systematic Literature Review
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-1EDDEA256F85	2025-10-16	openalex.org	1	Evaluation of computer vision techniques for quality inspection in casting manufacturing process
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-66E8F1ED640C	2025-09-18	openalex.org	1	Textile Defect Detection Using Artificial Intelligence and Computer Vision—A Preliminary Deep L...
SIG-D46EBCBB2C04	2025-09-16	openalex.org	1	Embedded vision transformers for industrial welding defect detection
SIG-D44EEFF68BEA	2026-08-13	manilatimes.net	2	Odysight . ai signs agreement with Boeing to demonstrate AI - Powered capabilities
SIG-5ADA6B03AA90	2026-05-05	arxiv.org	3	Physics-Grounded Multi-Agent Architecture for Traceable, Risk-Aware Human-AI Decision Support i...
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: `what_is_changing` (named entities that were never supplied: Boeing, Odysight.ai); diagram (4 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:13+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.